

ThunderClad 3

TU-933

Super Low Loss Material

tuc 台燿科技

Taiwan Union Technology Corporation (TUC)

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台灣新竹廠 · 江苏常熟廠 · 广东中山廠
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- Laminates & Prepregs
- Mass Lamination Service
- Insulated Metal Substrate Materials



TUC Product Roadmap

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2



ThunderClad 3



Features :

- ❑ Non-epoxy resin system
- ❑ For high speed digital / RF application
- ❑ Thermal robust for fine pitch design
- ❑ Flat Dk / Df varied with frequency & temperature
- ❑ $Dk = 3.4 @ 10GHz$ (split post cavity method)
- ❑ $Df = 0.0025 @ 10GHz$ (split post cavity method)
- ❑ $Tg = 170^{\circ}C / 220^{\circ}C$ (TMA / DMA)

Status : alpha site internal evaluation

ThunderClad 3



Type	ThunderClad 3 (TU-933)	TU-872 SLK	4350B*
<u>Performance</u>			
Tg, (DMA), °C	220	220	-
Tg, (TMA), °C	170	190	-
Td, (TGA, 5%wt loss) °C	390	340	390
Z-axis Expansion, (50-260 °C), %	2.7	2.3	1.1
Dk @ 10GHz (SPC method)	3.4	3.8	3.48
Df @ 10GHz (SPC method)	0.0025	0.0090	0.0037
Flammability, UL94	94-V0	94-V0	94-V0
Peel strength (1oz), lb/in	4~6	5~7	5
T-260, min	>60	60	>60
T-288, min	>60	20	>60
T-300, min	>60	-	-
PCT/1hr/Dip (288 °C), sec	>120	>120	-

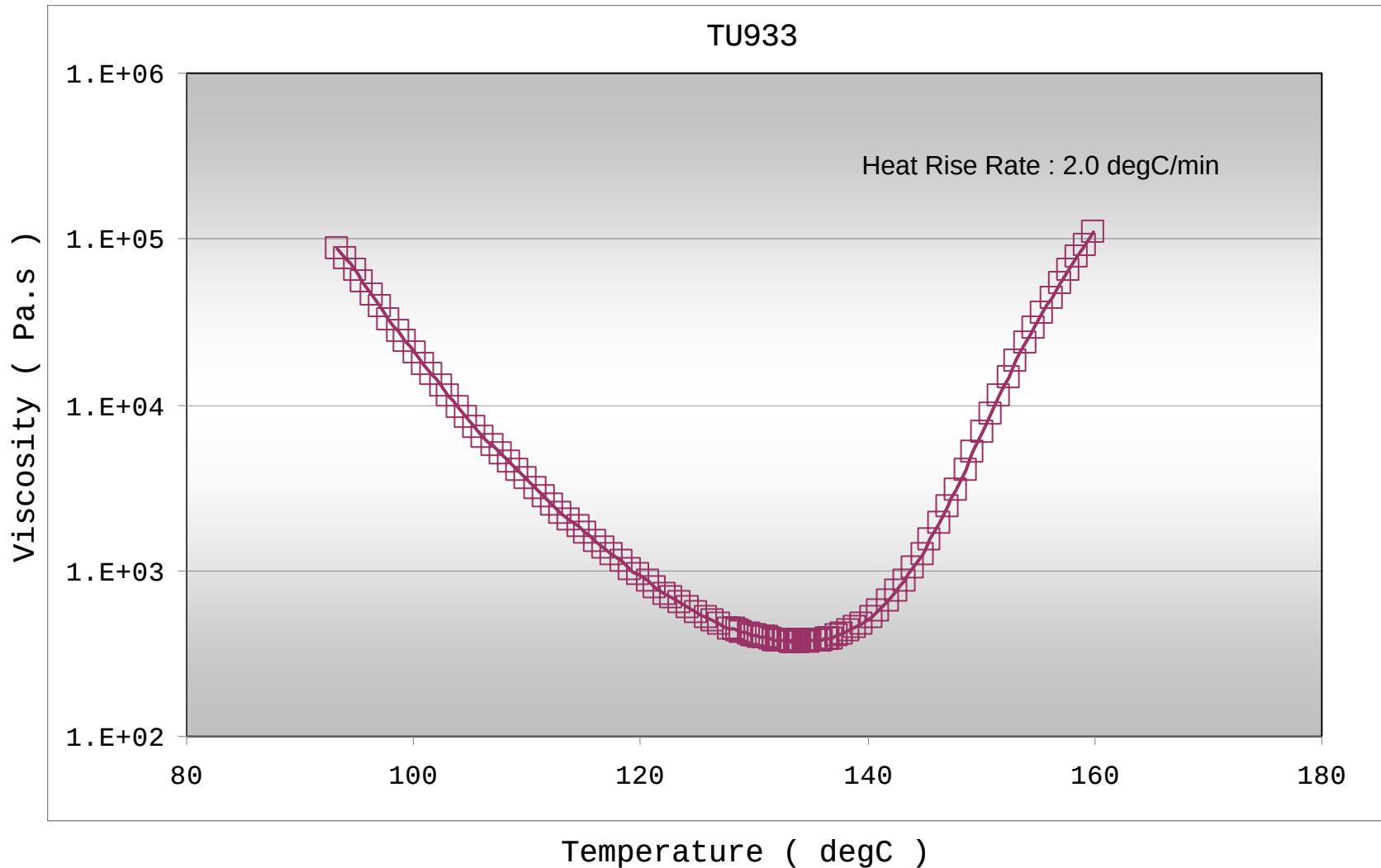
* Data is base on public datasheet information

ThunderClad 3 – Rheology

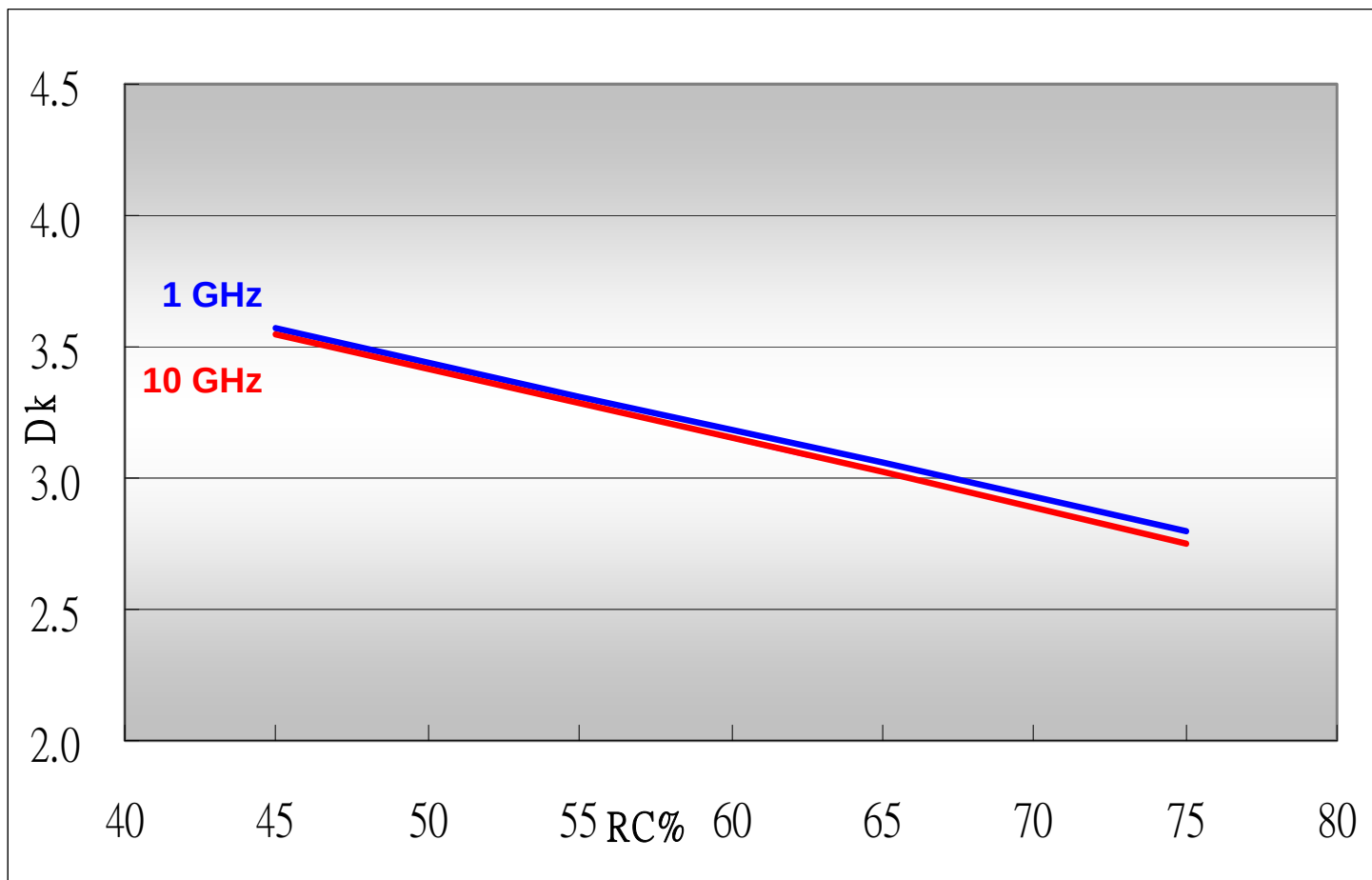
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5

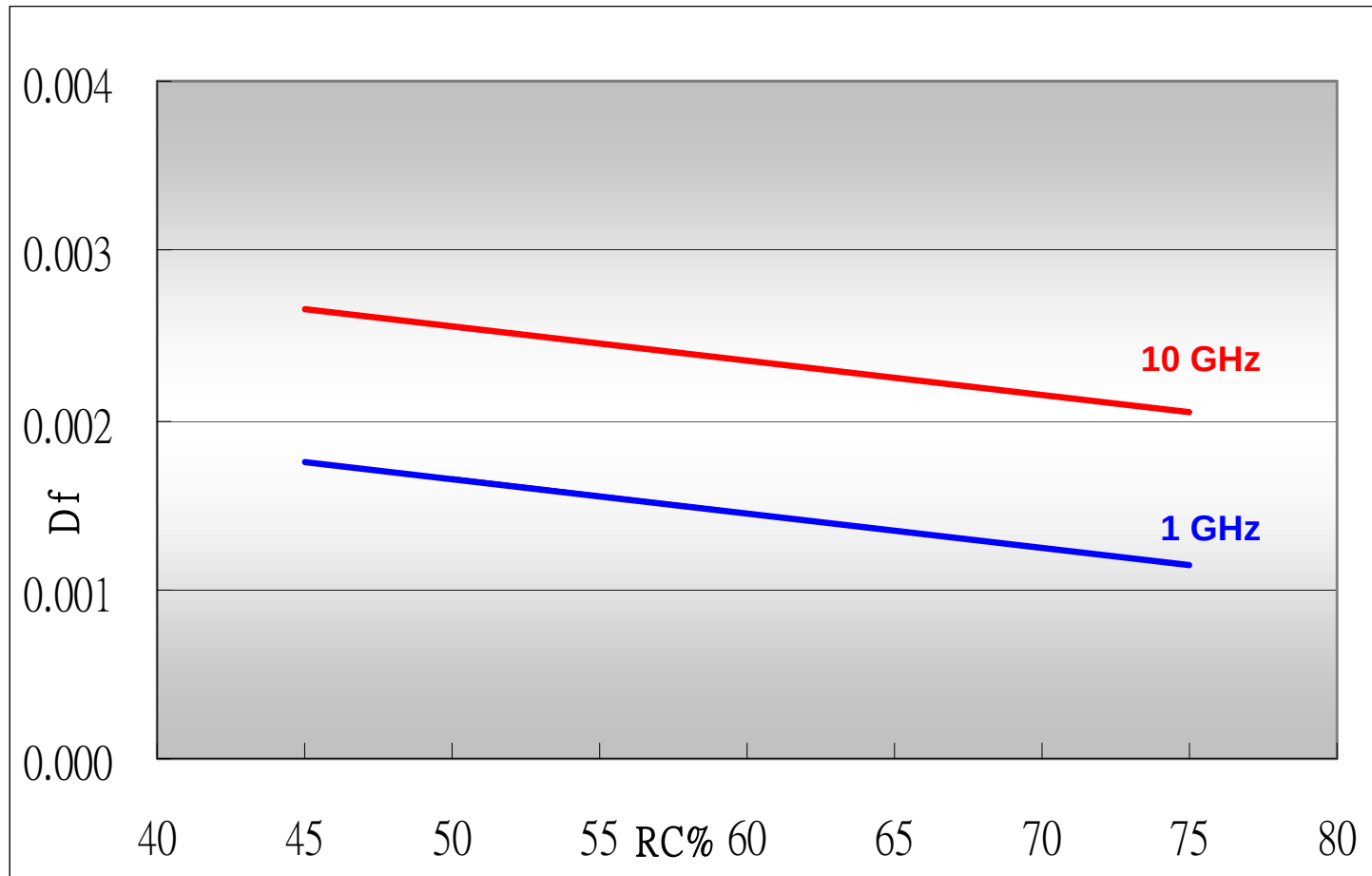


ThunderClad 3 – Dk



* Base on Split post Cavity Method

ThunderClad 3 – Df



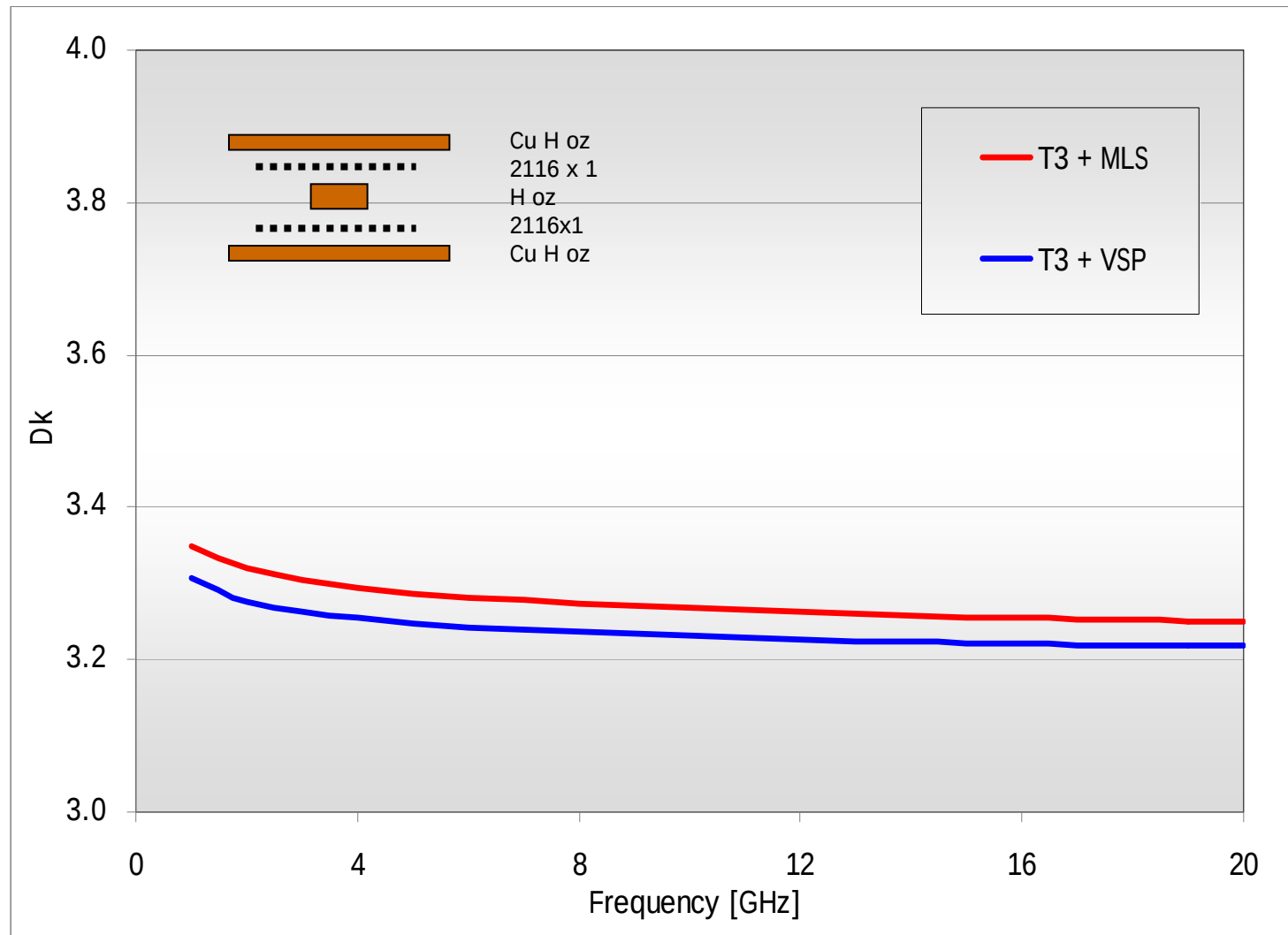
* Base on Split post Cavity Method

ThunderClad 3 – Propagation Constant

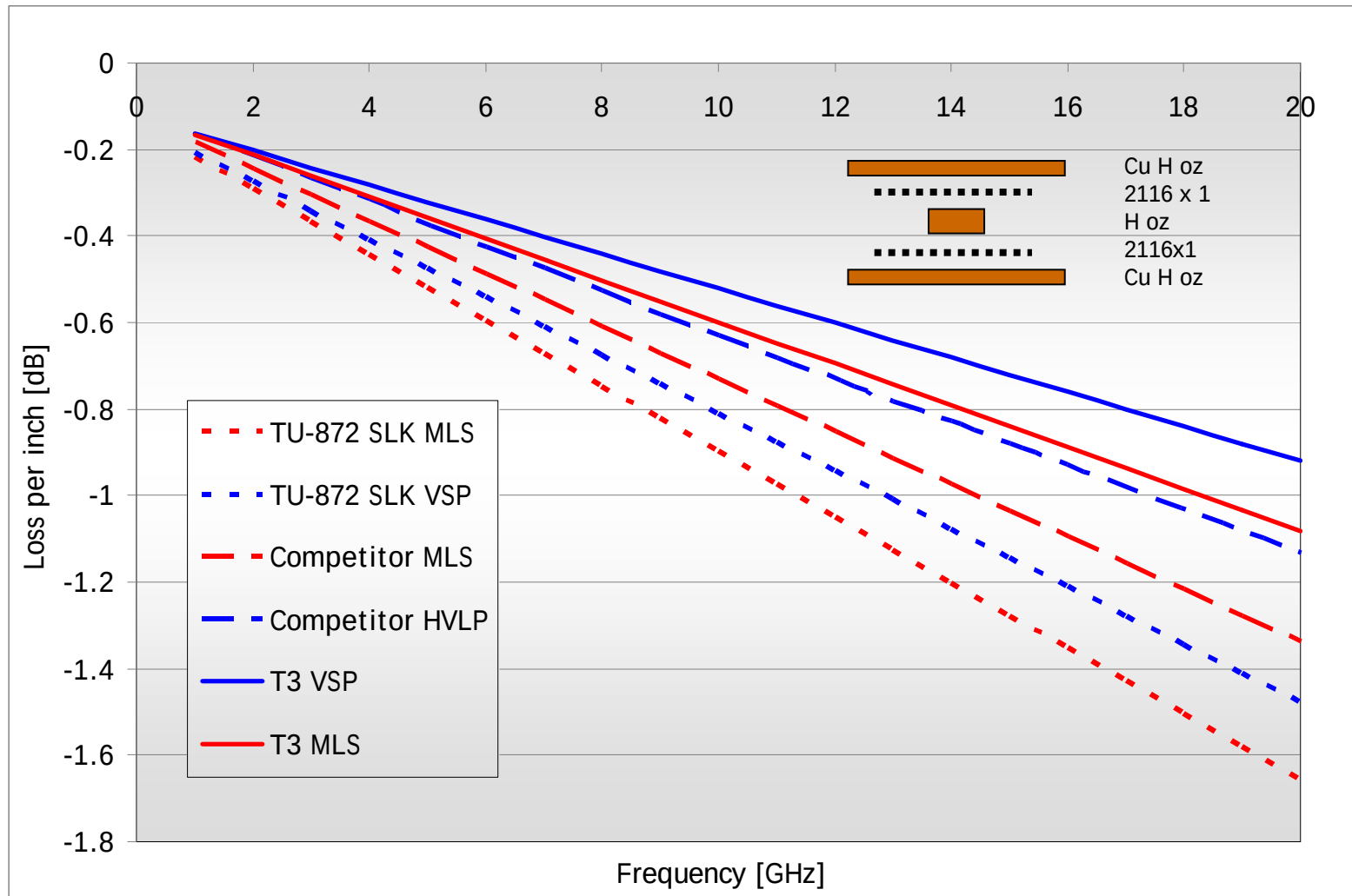
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8

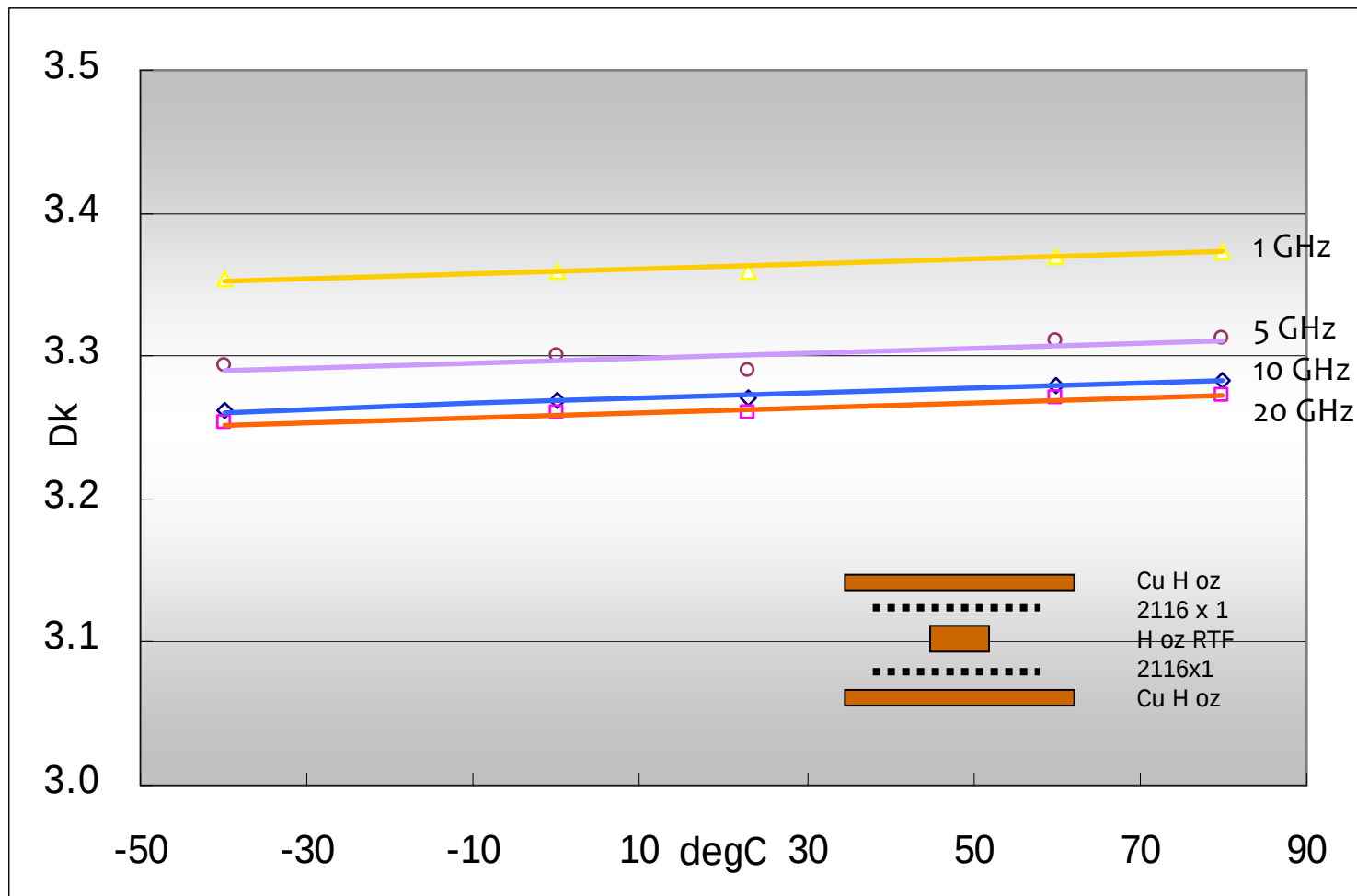


ThunderClad 3 – Insertion Loss



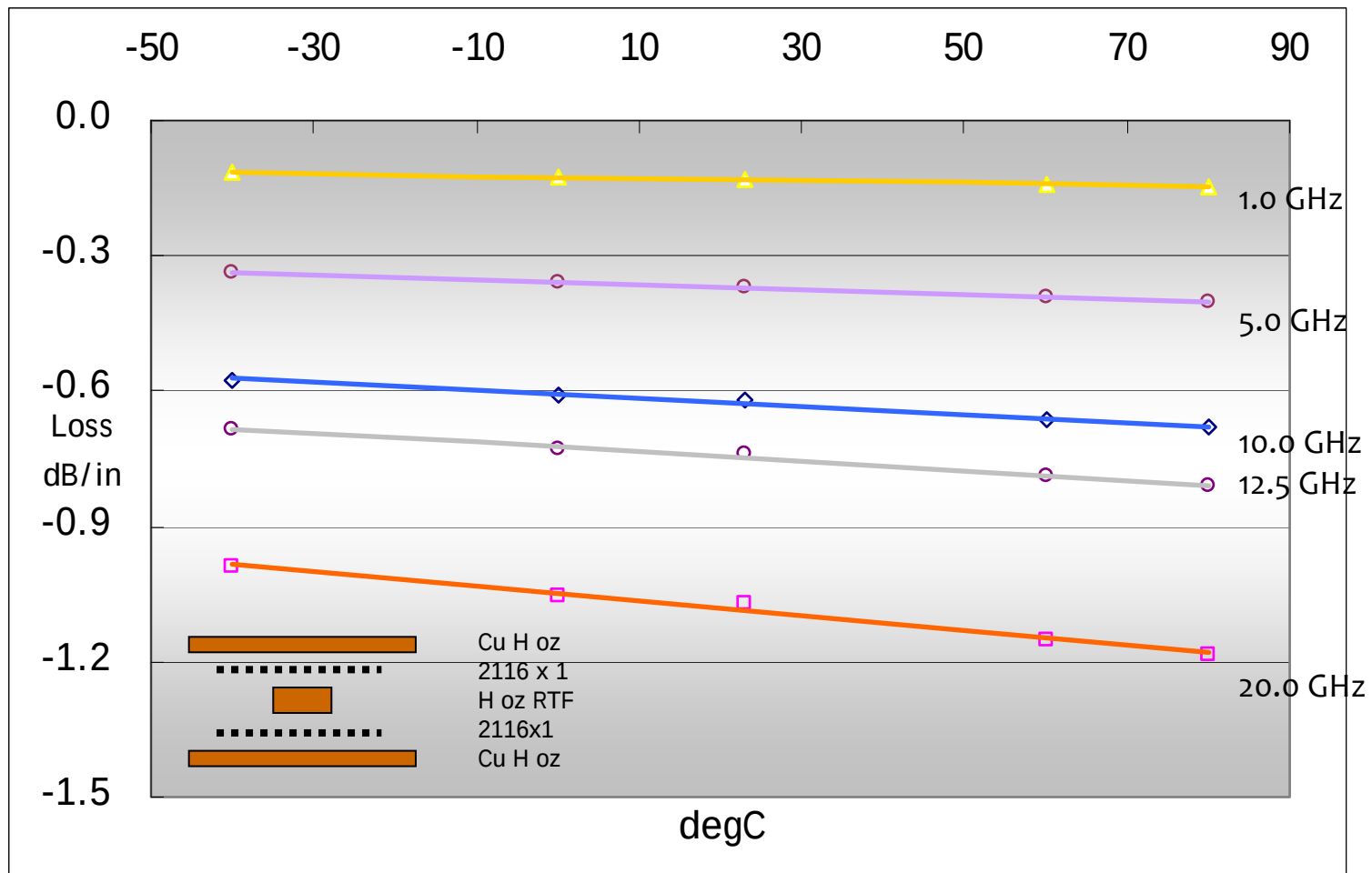
ThunderClad 3 – Dk vs Temperature

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ThunderClad 3 – Loss vs Temperature

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ThunderClad 3 – Case Study

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12

Material : ThunderClad 3

Layer count : 26 layer

Board Thickness : 3.2 mm

2 oz copper foil : x 4 layers

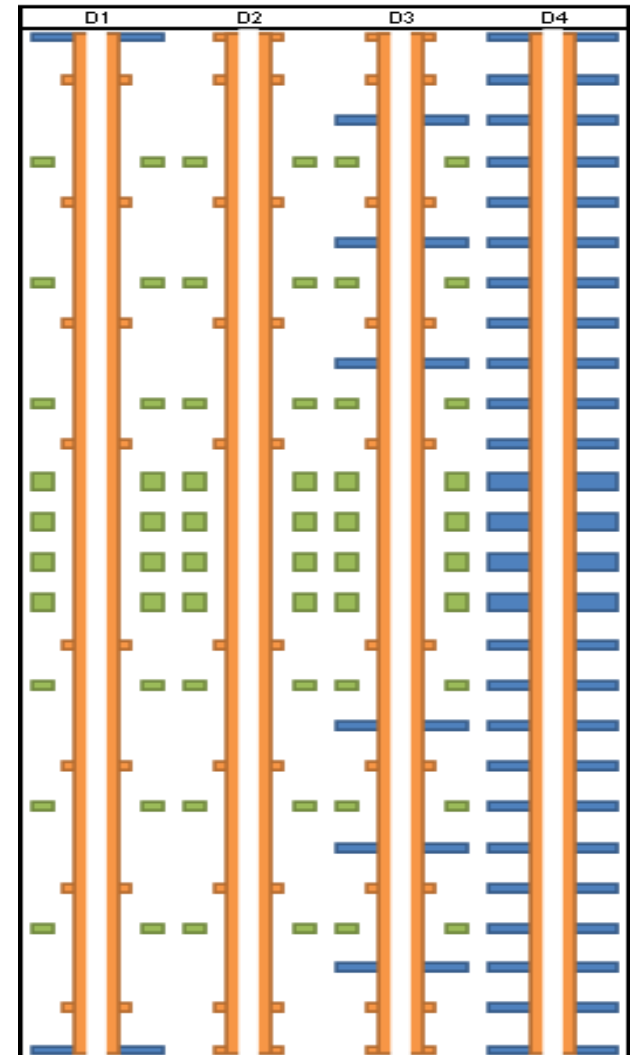
Pitch design : 0.65, 0.8 & 1.0 mm

Drilling hole size : 0.3 mm

Pattern design : x 4 types

Test criteria : 260°C reflow 10X

288°C Solder 6x



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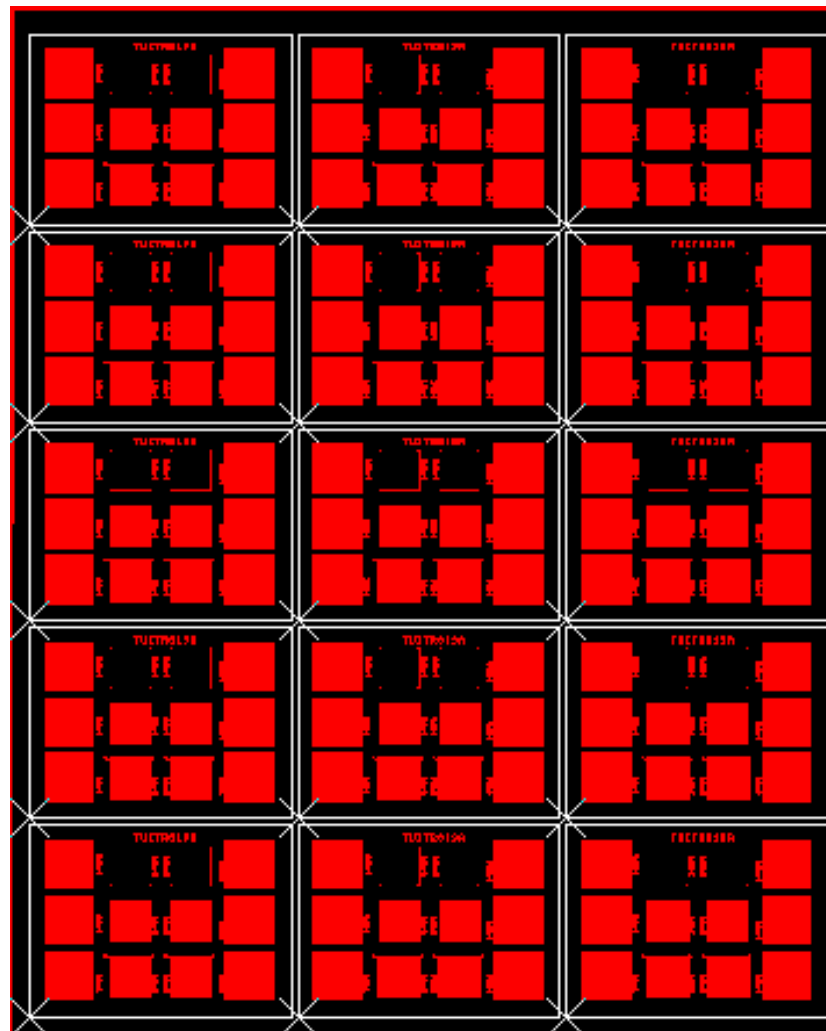
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13

	L1	H oz
		PP 2116 HRC × 1
	L2/3	3mil 1/1
		PP 2116 HRC × 1
	L4/5	3mil 1/1
		PP 2116 HRC × 1
	L6/7	3mil 1/1
		PP 2116 HRC × 1
	L8/9	3mil 1/1
		PP 2116 HRC × 1
	L10/11	3mil 1/1
		PP 106 HRC × 2
	L12/13	6mil 2/2 (2ply)
		PP 106 HRC × 3
	L14/15	6mil 2/2 (2ply)
		PP 106 HRC × 2
	L16/17	3mil 1/1
		PP 2116 HRC × 1
	L18/19	3mil 1/1
		PP 2116 HRC × 1
	L20/21	3mil 1/1
		PP 2116 HRC × 1
	L22/23	3mil 1/1
		PP 2116 HRC × 1
	L24/25	3mil 1/1
		PP 2116 HRC × 1
	L26	H oz



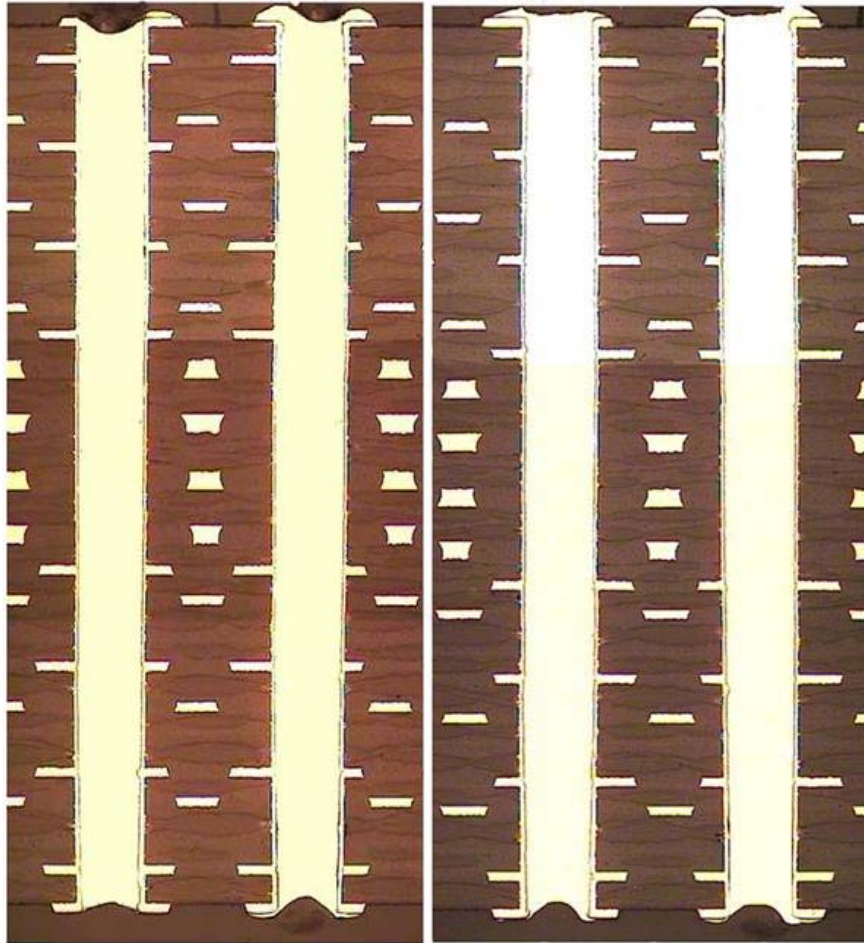
ThunderClad 3 – 0.8 mm / solder 6x

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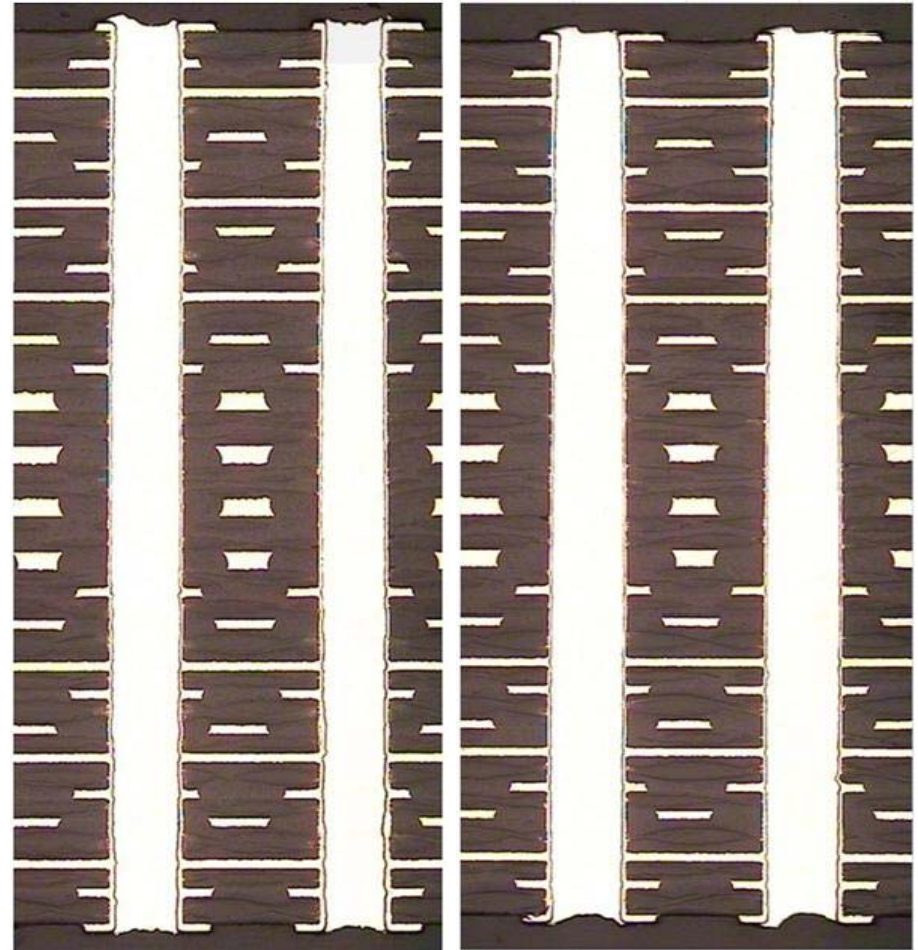
14

Solder 6x



D2 design

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D3 design

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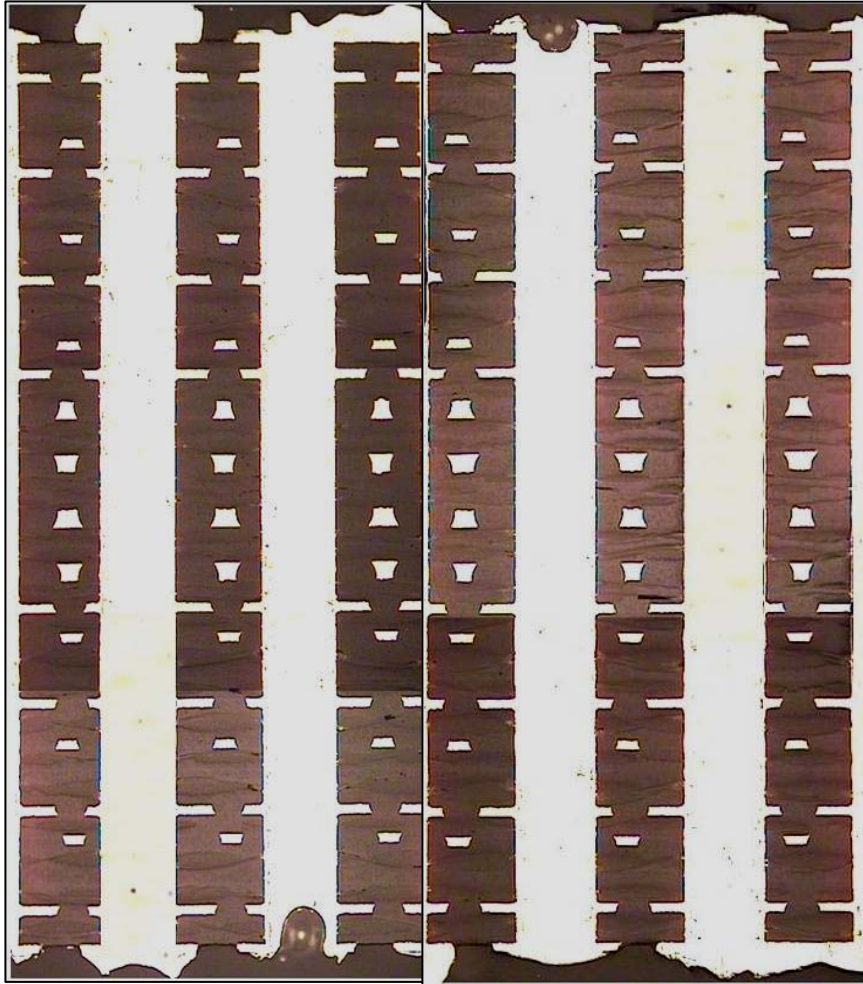
ThunderClad 3 – 0.65 mm / solder 6x

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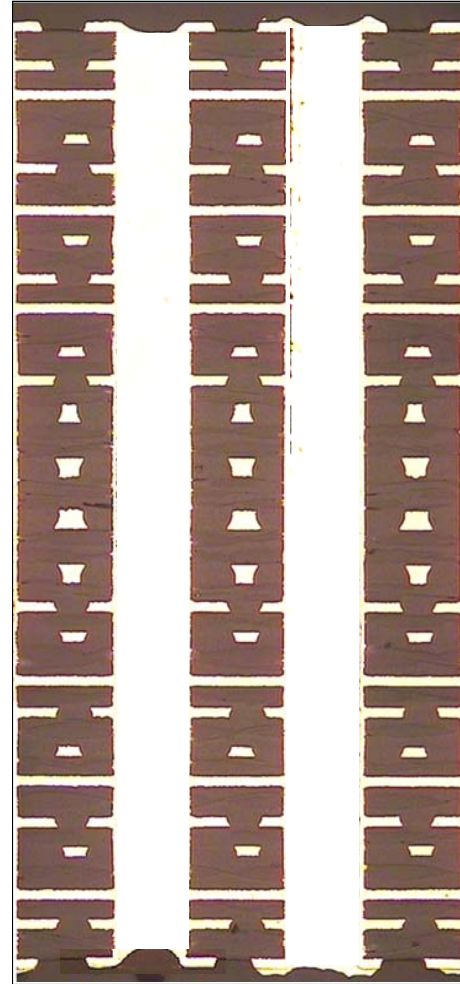


15

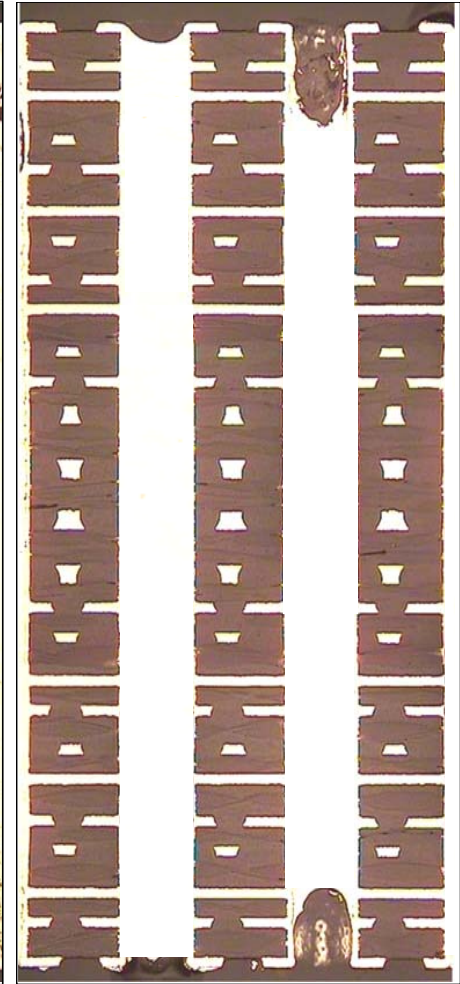
Solder 6x



D2 design



D3 design



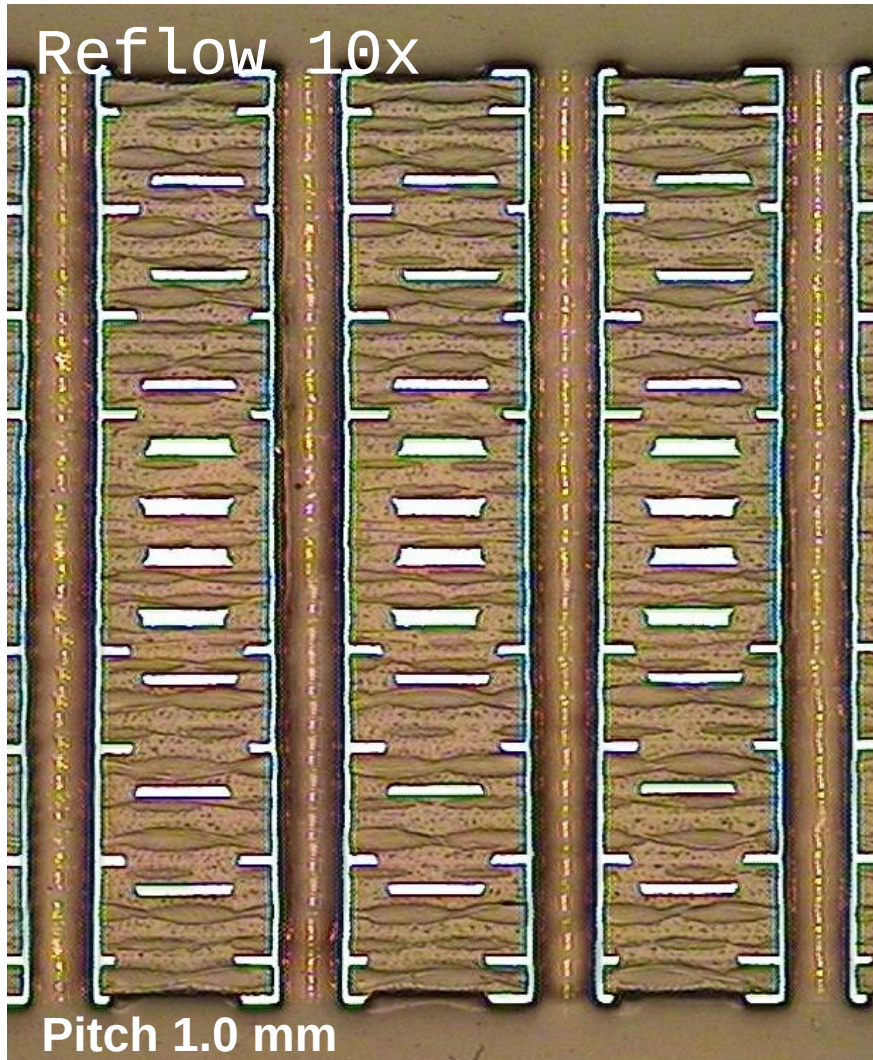
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ThunderClad 3 – 1.0 mm / reflow 10x

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16

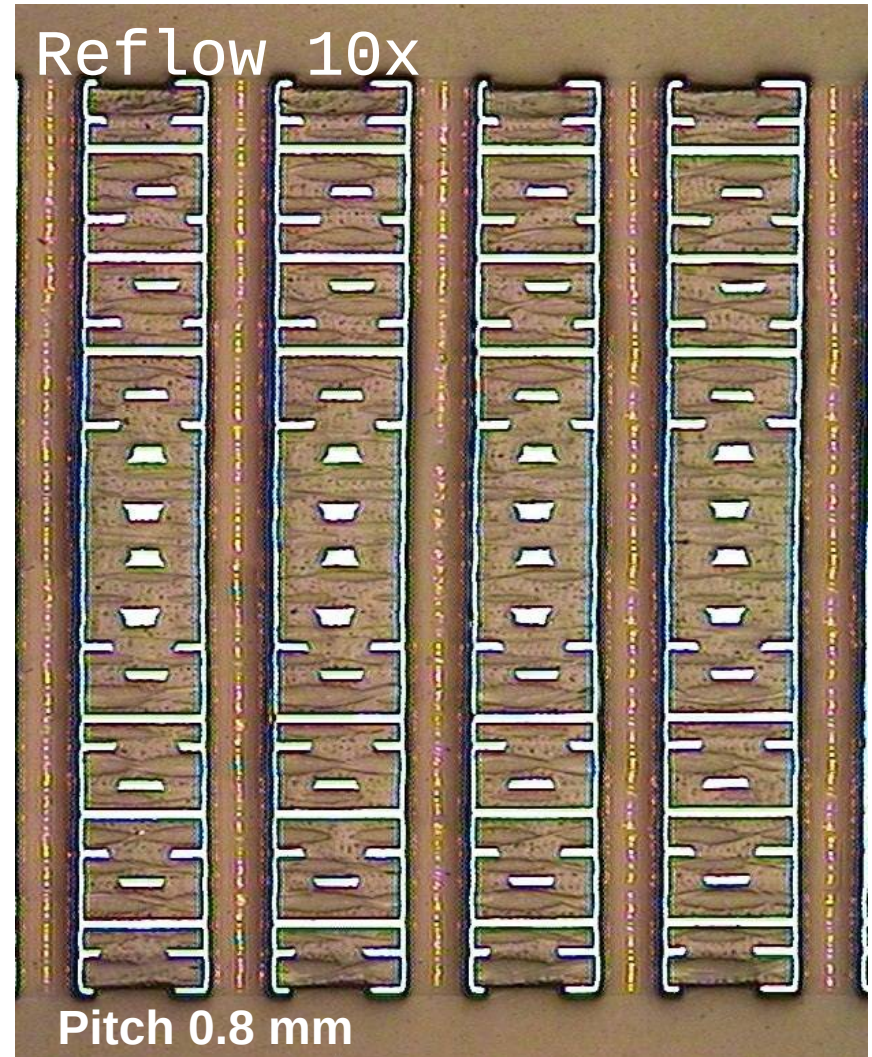
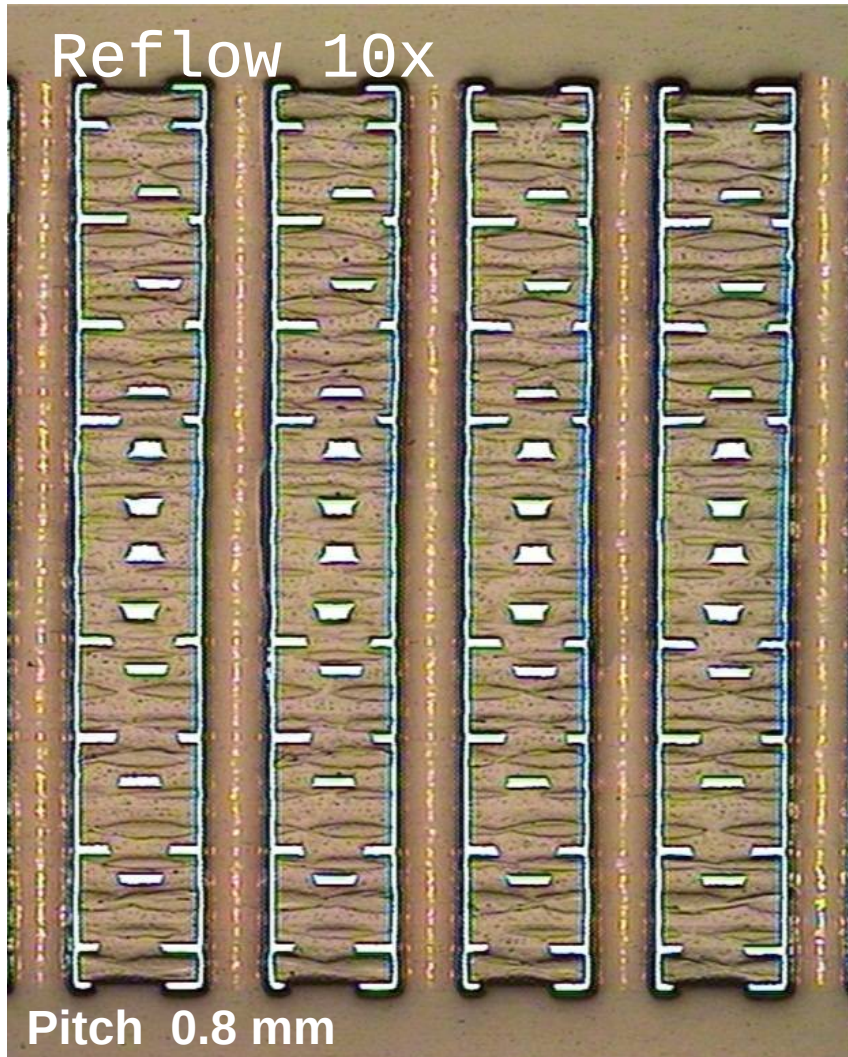


ThunderClad 3 – 0.8 mm / reflow 10x

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17

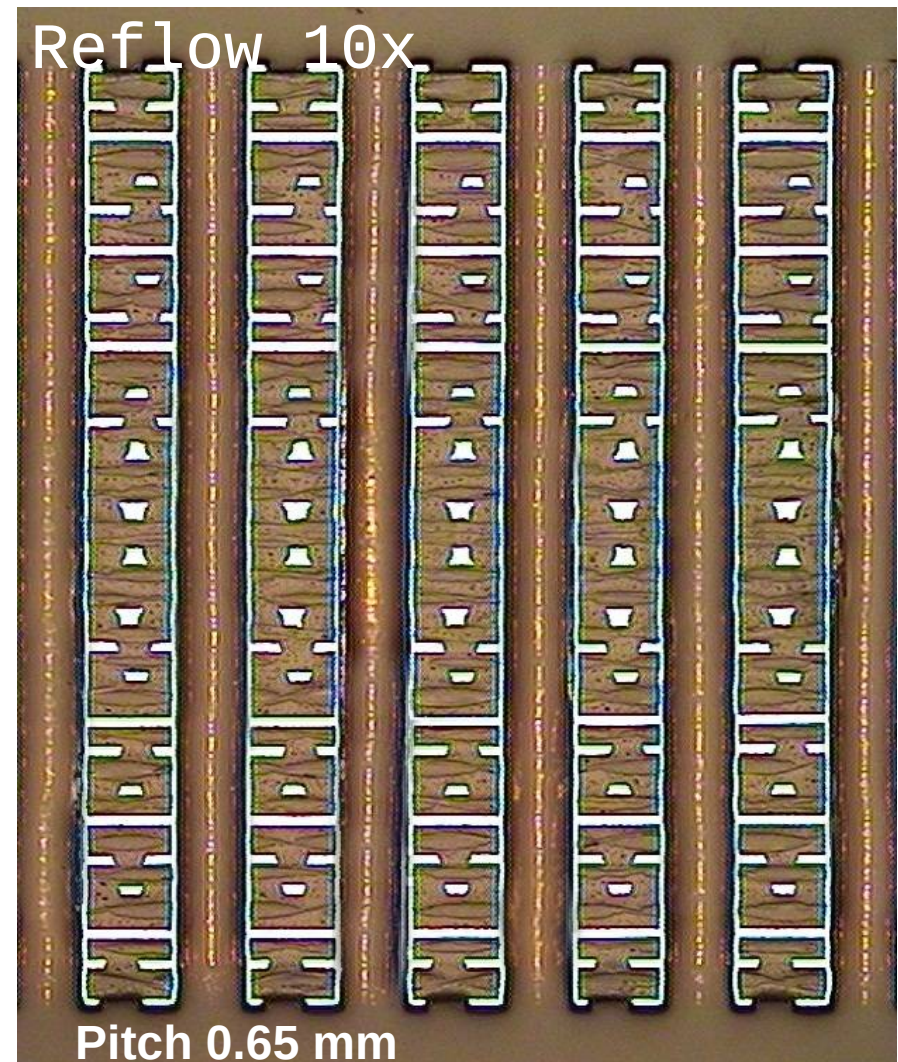
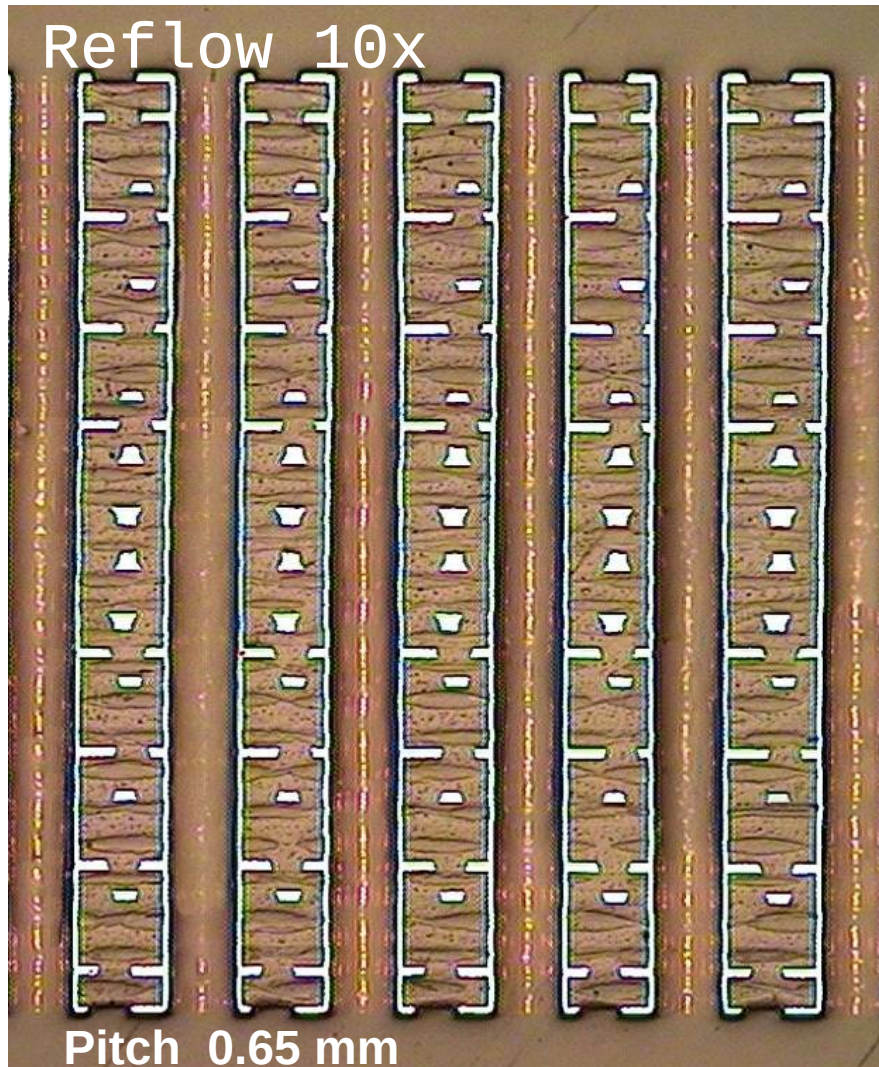


ThunderClad 3 – 0.65 mm / reflow 10x

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18



ThunderClad 3 – Case Study



Design / Thermal	Pitch 1.0 mm				Pitch 0.8 mm				Pitch 0.65 mm			
	D1	D2	D3	D4	D1	D2	D3	D4	D1	D2	D3	D4
Solder 6x	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Reflow 10X	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass

- No delamination
- No crack
- No ICD
- No Pad Lifting



Q & A